

CS232 Midterm

Problem Set Solution

P1.a

- (a) Throughput = $\min (R_i, i=1, 2, \dots, k)$;
- (b) μ links: $\min (R_i, i=1, 2, \dots, k)/\mu$;
- (c) $\sum_{n=1} L/(R_i/\mu) = \mu L \sum_{n=1}^k 1/R_i$;
- (d) L_2/R where $R = \min (R_i, i=1, 2, \dots, k)$;

P1.b

- (a) $L/R_1 + L/R_2 + d_1 + d_2$;
- (b) $L/R_1 + d_1 + (N-1)L/R_1 + L/R_2 + d_2 + (N-1)(L/R_2 - L/R_1)$ based on the fact: $R_1 > R_2$;

P1.c

- (a) 3 sec
- (b) 5 sec

P1.d

- (a) $e^{-\lambda s}$
- (b) $e^{-(R/L)\tau}$

P2.a

- (a) $\lambda_1 / (\lambda_1 + \lambda_2)$;
- (b) $\mu > (\lambda_1 + \lambda_2)$;
- (c) $(1/(\lambda_1 + \lambda_2 + \mu)) + (\mu/(\lambda_1 + \lambda_2 + \mu) \cdot [(1/(\lambda_1 + \lambda_2)) + (1/\mu)]) + [(\lambda_1 + \lambda_2)/(\lambda_1 + \lambda_2 + \mu)] \cdot (2/\mu)$;

P2.b

- (a) Router 1: $\min (R_4, R_1/3, R_2/2)$;
Router 2: $\min (R_4, R_1/3, R_3)$;
Router 3: R_5 ;
 $\max (\text{Router1}, \text{Router2}, \text{Router3})$;
- (b) $R_5 + \min (R_4, R_1/3, R_2/2 + R_3)$;
- (c) Packet Size / $\max (\text{Router1}, \text{Router2}, \text{Router3})$; Q^n

P2.c

- (a) $1 - P(0) = 1 - 1 / \sum_{n=1}^3 (Q^n / n!) ;$
- (b) $\mu \sum_{k=0}^c kp(k) ;$
- (c) $\lambda / (2\mu + \lambda);$

P3.a

- (a) $1/\mu_1 + 1/\mu_2 ;$
- (b) $\mu_1 / (\mu_1 + \mu_2) = 1/2;$
- (c) $5/(4\mu) ;$
- (d) $9/(4\mu)$

P3.b

- (a) $1 - (1-p)^k = 1 - p^k ;$
- (b) $1/(k\mu);$
- (c) $(1/\mu) \cdot \sum_{n=1}^k 1/n ;$
- (d) $1 - p^k;$
- (e) $E[T1] = 1/(k\mu) , E[\text{replica}] = 1/\mu .$